Clitic Splitting: Plausible and Over-Segmented Examples

Real XLM-R SentencePiece pieces (the tokenizer of multilingual-e5 and bge-m3) vs. the single-suffix clitic heuristic, on the most frequent words it splits

A. Splits that match a genuine Sorani suffix

کوردستاندا

19 occurrences in the corpus — valid split

XLM-R pieces (4)	کو	رد	ستان	دا	
Clitic heuristic	کوردستان	+	دا		

کاتێکدا

16 occurrences in the corpus — valid split

XLM-R pieces (5)	کا	ت	ێ	ک	دا	
Clitic heuristic	کاتێک	+	دا			

تایبەتدا

19 occurrences in the corpus — valid split

XLM-R pieces (4)	تا	ی	بەت	دا	
Clitic heuristic	تایبەت	+	دا		

B. Over-segmentation: the word ends in a clitic-like string but is not inflected

کوردستان

446 occurrences in the corpus — over-segmentation

XLM-R pieces (3)	کو	رد	ستان		
Clitic heuristic	کوردس	+	تان		

بەغدا

59 occurrences in the corpus — over-segmentation

XLM-R pieces (4)	_	بە	غ	دا	
Clitic heuristic	بەغ	+	دا		

پەرلەمان

39 occurrences in the corpus — over-segmentation

XLM-R pieces (4)	_	پەر	لە	مان	
Clitic heuristic	پەرلە	+	مان		

MEASURED ON 71,900 WORD TOKENS (16,494 TYPES) OF THE 492 PASSAGES

Word tokens split by heuristic	Word types split	XLM-R pieces/word: raw → split	BM25 stems vs. word, ΔMRR@10 (95% CI)
6.65%	12.96%	3.354 → 3.381	+0.16 [-0.32, +0.67]

Words are the most frequent ones the heuristic splits; the valid/over-segmentation labels are the authors' judgement and have not been validated by an independent annotator. The splitter has no lexicon, so it cannot tell inflected words from words that merely end in the same letters.